OPTIMISATION

1. a) The first order necessary conditions of optimality are

$$0 = \frac{\partial f}{\partial x} = y(2x+y-1), \qquad 0 = \frac{\partial f}{\partial y} = x(x+2y-1),$$

yielding the stationary points

$$P_1 : (0,0), \quad P_2 : (0,1), \quad P_3 : (1,0), \quad P_4 : \left(\frac{1}{3}, \frac{1}{3}\right).$$

[4 marks]

b) Note that

$$\nabla^2 f = \begin{bmatrix} 2y & 2x+2y-1 \\ 2x+2y-1 & 2x \end{bmatrix}.$$

Hence P_1, P_2, and P_3, are saddle points, and P_4 is a minimizer. [4 marks]

c) Along the line $y = x$ one has $f(x,x) = x^2(2x-1)$, which is not bounded from below, nor from above. Hence the function f does not have a global minimizer/maximizer. [2 marks]

d) The gradient iteration is (note that x^+ denotes x_{k+1} and x denotes x_k, and similarly for y):

$$x^+ = -\alpha y(2x+y-1)+x, \qquad y^+ = -\alpha x(x+2y-1)+y.$$

Setting $\alpha = 1$ and $x_1 = y_1 = 1/3$, and solving for x and y yields

$$(x_0, y_0) = \left(-\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right), \qquad (x_0, y_0) = \left(\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right),$$

which are the only initial conditions from which the gradient iteration, with $\alpha = 1$, converges to the minimizer. [4 marks]

e) Rewriting the Newton's iteration as suggested, using $x_1 = y_1 = 1/3$, and solving for x and y yields

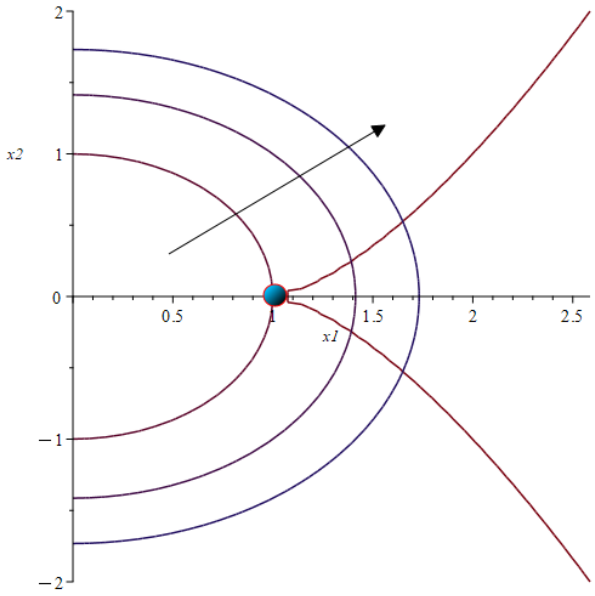
$$(x_0, y_0) = \left(\frac{1}{9}, \frac{1}{9}\right), \qquad (x_0, y_0) = \left(\frac{1}{9}, \frac{7}{9}\right), \qquad (x_0, y_0) = \left(\frac{7}{9}, \frac{1}{9}\right).$$

[2 marks]

f) The global minimizer is either at the boundary of the admissible set or it is a stationary point in the interior of the admissible set. Note that, on the boundary of the admissible set, the function f is equal to 0, whereas at the point P_4, which is the only stationary point inside the admissible set, $f(P_4) = -1/27$. Hence P_4 is the global minimizer of the considered problem. [4 marks]

Answers

2. a) The level lines of the cost function and the admissible set are sketched in the graph below. The function f increases along the direction of the arrow.



[4 marks]

b) Note that

$$\frac{\partial g}{\partial x} = \begin{bmatrix} 3(x_1-1)^2 & -2x_2 \end{bmatrix}.$$

As a result, all points of the admissible set are regular, with the exception of the point $(1,0)$. [2 marks]

c) By inspection of the sketch of the level lines of the cost and of the admissible set one concludes that the optimal point is $(1,0)$ (the blue dot in the plot), giving the optimal cost $f = 1$. [2 marks]

d) The Lagrangian for this problem is

$$L(x_1,x_2,\lambda) = x_1^2 + x_2^2 + \lambda((x_1-1)^3 - x_2^2).$$

The necessary conditions of optimality are

$$0 = 2x_1 + 3\lambda(x_1-1)^2, \qquad 0 = -2x_2(\lambda-1), \qquad 0 = (x_1-1)^3 - x_2^2.$$

These do not return any candidate optimal point. This is because the optimal solution of the problem is a non-regular point. [4 marks]

e) Replacing $x_2^2 = (x_1-1)^3$ in the cost yields

$$f_{\text{red}}(x_1) = x_1^2 + (x_1-1)^3.$$

This function is not bounded from below, hence the problem obtained by eliminating the constraint in this way does not have a solution. This is the case because the constraint elimination procedure implemented above does not consider that x_2^2 is non-negative, hence does not carry over the condition $x_1 \geq 1$. [4 marks]

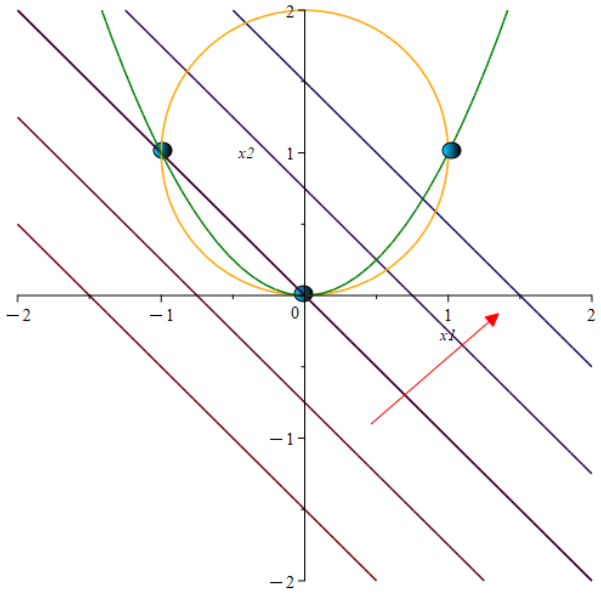
A properly reduced problem is

$$\min_{x_1} x_1^2 + (x_1-1)^3, \qquad x_1 \geq 1.$$

The solution of this problem is $x_1 = 1$, which coincides with the solution determined in part c). [4 marks]

Answers

3. a) The level lines of the cost function and the admissible set are sketched in the graph below. Note that the admissible set is composed of the three points indicated with large dots. The function f increases along the direction of the arrow.



The optimal solution of the problem is achieved at $P_0 = (0,0)$ and $P_1 = (-1,1)$. For both points the cost is $f = 0$. Note that at $P_2 = (1,1)$, $f = 2$, and this is the global maximizer. [4 marks]

b) A direct computation shows that $(0,0)$ is not a regular point for the constraints, whereas $(-1,1)$ and $(1,1)$ are regular points. [2 marks]

c) The Lagrangian for the problem is

$$L(x_1,x_2,\lambda_1,\lambda_2) = x_1 + x_2 + \lambda_1(x_1^2 + (x_2-1)^2 - 1) + \lambda_2(x_1^2 - x_2).$$

The necessary conditions of optimality are

$$0 = 2\lambda_1 x_1 + 2\lambda_2 x_1 + 1, \qquad 0 = (2x_2 - 2)\lambda_1 - \lambda_2 + 1,$$

$$0 = x_1^2 + (x_2-1)^2 - 1, \qquad 0 = x_1^2 - x_2.$$

Replacing P_0, P_1, and P_2 in the necessary conditions of optimality shows that for P_1 and P_2 there are multipliers λ_1 and λ_2 such that the necessary conditions are satisfied. At P_0, which is not a regular point, the conditions are not satisfied. [4 marks]

d) The augmented Lagrangian evaluated at P_1 is (note that $\lambda_1 = -1/2$ and $\lambda_2 = 1$)

$$L_a(x_1,x_2) = \frac{x_1^2}{2} - \frac{(x_2-1)^2}{2} + \frac{1}{2} + x_1 + \frac{(x_1^2 + (x_2-1)^2 - 1)^2 + (x_1^2 - x_2)^2}{\varepsilon}.$$

[2 marks]

Note now that

$$\frac{\partial L_a}{\partial x_1} = x_1 + 1 + \frac{4(x_1^2 + (x_2-1)^2 - 1)x_1 + 4(x_1^2 - x_2)x_1}{\varepsilon},$$

$$\frac{\partial L_a}{\partial x_2} = -x_2 + 1 + \frac{2(x_1^2 + (x_2-1)^2 - 1)(2x_2 - 2) - 2x_1^2 + 2x_2}{\varepsilon},$$

Answers

and that

$$\frac{\partial L_a}{\partial x_1}(P_1) = \frac{\partial L_a}{\partial x_2}(P_1) = 0,$$

that is P_1 is a stationary point of L_a. [4 marks]

Finally,

$$\nabla^2 L_a(P_1) = \begin{bmatrix} 1+\dfrac{16}{\varepsilon} & \dfrac{4}{\varepsilon} \\ \dfrac{4}{\varepsilon} & -1+\dfrac{2}{\varepsilon} \end{bmatrix}$$

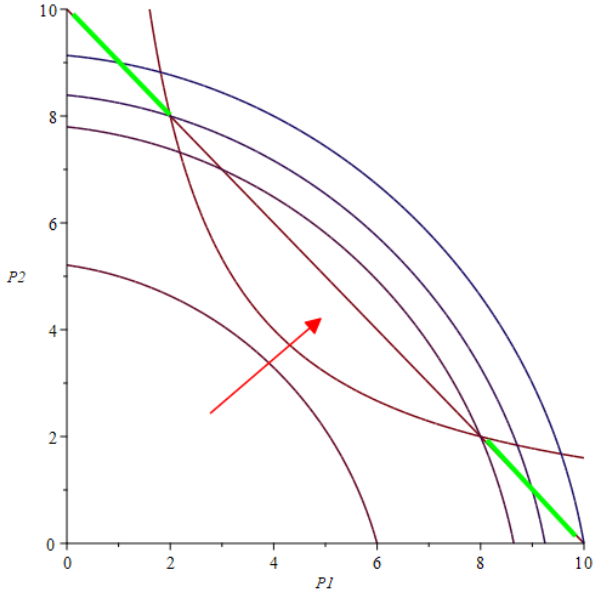
and

$$\det\nabla^2 L_a(P_1) = \frac{16-14\varepsilon-\varepsilon^2}{\varepsilon^2},$$

which shows that $\nabla^2 L_a(P_1)$ is positive definite, hence P_1 is a local strict minimizer, for all $\varepsilon \in (0, \sqrt{65}-7)$. [4 marks]

Answers

4. a) The level lines of the cost function and the admissible set (in green) are sketched in the graph below. Note that the admissible set is composed of two segments. All points are regular points. The function f increases along the direction of the red arrow.



The optimal solution is achieved at $P_1 = 8$ and $P_2 = 2$. [6 marks]

b) The Lagrangian of the problem is

$$L(P_1, P_2, \lambda, \mu) = 2 + 2P_1 + 4P_2 + P_1^2 + P_2^2 + \lambda(P_1 + P_2 - 10) + \mu(P_1 P_2 - 16).$$

The necessary conditions of optimality are

$$0 = 2 + 2P_1 + \lambda + \mu P_2, \qquad 0 = 4 + 2P_2 + \lambda + \mu P_1,$$
$$P_1 + P_2 = 10, \qquad P_1 P_2 - 16 \leq 0, \qquad \mu \geq 0, \qquad \mu(P_1 P_2 - 16) = 0.$$

[2 marks]

c) The candidate optimal points are

$$P_1 = 2, \;\; P_2 = 8, \;\; \lambda = -74/3, \;\; \mu = 7/3,$$

and

$$P_1 = 8, \;\; P_2 = 2, \;\; \lambda = -64/3, \;\; \mu = 5/3.$$

[6 marks]

d) Note that at the candidate optimal points there are two active constraints, hence there is no non-zero vector s which is in the nullspace of the Jacobian of the active constraints. [2 marks]

Eliminating the equality constraint yields the problem

$$\min_{P_1} 2P_1^2 - 22P_1 + 142,$$
$$P_1(10 - P_1) \leq 16.$$

Note that the inequality constraint is equivalent to say that $P_1 \in [0,\ 2] \cup [8,\ 10]$. Evaluating the reduced cost f_{red} at the boundary of the admissible set (note that the reduced cost does not have any stationary point in the admissible set) yields

$$f_{red}(0) = 142, \qquad f_{red}(2) = 106, \qquad f_{red}(8) = 94, \qquad f_{red}(10) = 122.$$

Hence, the (global) minimizer for the considered problem is $P_1 = 8$ and $P_2 = 2$. [4 marks]

Answers